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Shipment of insects and related arthropods into and out of India for research or commercial purposes

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Summary

This paper provides a concise summary of the guidelines and regulations that govern the shipment of insects and related arthropods into and out of India. It discusses the Plant Quarantine (Regulation of Import into India) Order, 2003, which regulates the import of agents for biological control, and the Biological Diversity Act, 2002, and associated Biological Diversity Rules, 2004, which guide the transfer of insects and related arthropods for identification, taxonomic research, and commercial purposes. Insect trade for some food and feed purposes is also mentioned. Finally, some implications, conclusions and recommendations are presented.

Keywords

Biocontrol agent – Bioresource – India – Insect exchange – National Biodiversity Authority.

Introduction

The documented history of insect shipments to India dates back to the 18th century. These first arrivals were all related to the efforts of the East India Company to introduce the 'true cochineal' insect (*Dactylopius coccus*) (Hemiptera: Dactylopiidae) into India for the dye industry. Even though there was no real success in this endeavour, the narrative on the attempts at establishing a lucrative cochineal trade makes interesting reading. Tryon [1] gives an account of these efforts, citing reports by East India Company official G.A. Prinsep from 1838/9. He also refers to the works of Whitelaw Ainslie [2], Superintending Surgeon of the then Madras Establishment, and William Roxburgh [3], the then Superintendent of the Botanical Gardens in Calcutta (now Kolkata). The first consignment of the insects was hand-carried from Brazil by Captain Richard Neilson and handed over to Roxburgh in 1795, and imports continued into the early 19th century. However, the insects imported had been mistakenly identified: they were not, in fact, *D. coccus* ('grana fina'), but *D. ceylonicus* ('grana silvestre'), which produces an inferior dye. Luckily, however, the latter turned out to be valuable as a biocontrol agent for prickly pear cacti, particularly *Opuntia monacantha* (Cactaceae).

The difficulties of importing live insects are exemplified by the experience of the then Mysore Department of Agriculture, which in the early 1900s attempted to introduce the seed fly *Ophiomyia lantanae* (= *Agromyza lantanae*) (Diptera: Agromyzidae) from Hawaii on five different occasions. Most of the insects died in transit, almost leading to the failure of the biocontrol programme for which they were intended, namely, the programme to control *Lantana camara* (Verbenaceae), the lantana weed [4]. India has come a long way since then, and robust regulations and guidelines are now in place for the shipment of insects and related arthropods into and out of India for research or commercial purposes.

Shipment of insects and related arthropods for biological control

The volume of insects imported increased dramatically from the 1960s onwards following the establishment of the Indian

office of the Commonwealth Institute of Biological Control in Bangalore (now Bengaluru) in 1957. Exotic insects for biocontrol purposes were prioritised and, following a period of quarantine, those considered safe to non-targets were released into the field as classical biocontrol agents. Similarly, most insects shipped out of India were exported primarily for biocontrol purposes. Thus, India played a significant role in the global exchange of insects, and other natural enemies, for biological control. In 1977, the Indian Council of Agricultural Research (ICAR) initiated the All-India Coordinated Research Project on Biological Control of Crop Pests and Weeds, which paved the way to further strengthen the importation of natural enemies for biological control of invasive insects and weeds. Since 1993, the importation and introduction of biocontrol agents, particularly insects, has been handled by the ICAR–National Bureau of Agricultural Insect Resources (ICAR–NBAIR) (known initially as the Project Directorate of Biological Control and, subsequently, as the National Bureau of Agriculturally Important Insects). Supporting the safe introduction of biocontrol agents is a key responsibility of this Bureau. It develops and/or oversees arrival procedures and quarantine regulations and provides comments on every importation application.

Plant Quarantine (Regulation of Import into India) Order, 2003

The Directorate of Plant Protection, Quarantine and Storage, which came into being in 1946, headed by the Plant Protection Advisor, is part of the Indian Government's Department of Agriculture, Cooperation and Farmers' Welfare in the Ministry of Agriculture and Farmers' Welfare. One of its mandates is to prevent the introduction of exotic pests inimical to Indian agriculture by implementing the Destructive Insects and Pests (Amendment and Validation) Act, 1992 (the Destructive Insects and Pests Act, 1914, is the principal Act). The implementation of this Act is supported by the Plant Quarantine (Regulation of Import into India) Order (PQO), 2003. The Order regulates the import and prohibition of import of plants and plant products into India. It contains specific requirements for permits for live-insect imports where they pertain to agriculture. It was published in the Gazette of India (S.O. 1322 [E], dated 18 November 2003) by the then Ministry of Agriculture, and has subsequently been amended 90 times, the last being on 10 June 2021 (an updated and consolidated version of the order is available at: http://ppqs.gov.in/archive-acts). The PQO, which came into effect on 1 April 2004, has 15 clauses that describe various aspects of the importation of agricultural articles (plants and plant products) and stipulate the conditions of their import into India. There are 16 administrative forms for various plant quarantine regulatory functions, two of which (Forms 12 and 13) relate specifically to the import of live insects [5]. The earlier order, 'Rules for regulating the import of insects into India', dated 3 February 1941, was repealed after the enactment of the PQO.

Plant Quarantine Order Chapter II: general conditions for import

As the primary aim of the PQO is to prevent the introduction of potentially harmful contaminant organisms along with colonies of potentially beneficial insects, the Order requires a taxonomic declaration and requires approved quarantine to ensure clean colonies. Paragraph 7 deals with the import of live insects and other arthropods/nematodes/microbial cultures, including algae/biocontrol agents. It reads as follows:

(1) No consignment of live insects and other arthropods/nematodes/microbial cultures including algae/biocontrol agents shall be permitted into India without valid import permit issued by competent authority as specified under Schedule-X.

(2) Every application or online application for permit to import live insects and other arthropods/nematodes/microbial cultures including algae/biocontrol agents, shall be made in the PQ Form 12 at least thirty days in advance to Plant Protection Advisor along with a fee of Rs 1,000 towards registration in the form of bank draft issued in favour of the Accounts Officer, Directorate of Plant Protection, Quarantine and Storage, Faridabad.

(3) The competent authority shall issue the permit in PQ Form 13 in triplicate, if satisfied of the purpose for which import is made and subject to such conditions imposed thereon.

(4) All the consignments of live insects and other arthropods/nematodes/microbial cultures, including algae/biocontrol agents, shall be permitted only through points of entry specified under Clause 3(14). The consignment of beneficial insects shall be accompanied by a certificate issued by National Plant Protection Organisation at the country of origin with additional declarations for freedom from specified parasites and parasitoids and the biocontrol agents free from hyperparasites. The consignment of beneficial insects/biocontrol agents shall be subject to post-entry quarantine as may be prescribed by the Plant Protection Advisor.

(5) Nothing contained in the clause shall apply to import of live insects and other arthropods/nematodes/microbial cultures, including algae/biocontrol agents, having no relevance in agriculture.

Quarantine facility for biocontrol agents

An exclusive quarantine facility is available at ICAR–NBAIR to handle imported arthropod and microbial biocontrol agents. It includes a quarantine area with HEPA (high-efficiency particulate air) filters and airtight doors, a glasshouse with a shatterproof polycarbonate roof, an incinerator, a double-ended autoclave, and other pieces of equipment needed to handle various imported organisms. Although this facility has been providing services exclusively for Indian biocontrol

programmes, it could also serve other countries in the region that lack quarantine facilities, if needed [6].

Regulations for export of live insects and related arthropods for biological control

Export shipments of live classical biocontrol agents that are effective against potential invasive pests in other countries are required to comply with guidelines published by the National Biodiversity Authority (NBA), which is an autonomous statutory body that falls under the responsibility of the Ministry of Environment, Forestry and Climate Change located in Chennai. It is responsible for regulating access to the country's biological resources and oversees the implementation of the Biological Diversity Act, 2002 (BDA), one of the objectives of which is the 'fair and equitable sharing of the benefits arising out of the use of biological resources'. The nature of the biological resources must be assessed to determine whether they are native to India or have been naturalised in India. If the insect species is indigenous to India, then its export is governed by the BDA and permission to export it must be obtained from the NBA [7]. If the insect species is an exotic biocontrol agent that was imported long ago for biological control and the species has been established in India for years, the BDA does not apply. In such cases, the country that needs the biocontrol agent can send a request letter directly to the institute supplying the insects. A committee of scientists at the institute prepares a proposal for export for each individual case and confirms/certifies that the requested biocontrol agent is exotic. Based on the committee's comments and recommendations, the director of the institute will decide if the insects can be exported. If the proposal is approved, the biocontrol agents can be exported for the purpose of non-commercial research or for use in a global biocontrol programme. The shipment will be accompanied by a certificate confirming the exotic nature of the biocontrol agents. In all cases, a voucher specimen or sample of the species should be deposited in the institute repository.

Packing and shipping of insects and related arthropods for biological control

Indian scientists follow international procedures and protocols for packing and shipping insects and related arthropods for biological control. Because of the involvement of at least two countries in such transfers, common protocols are a must. Early reviews, such as the one by Boldt and Drea [8], laid the foundation to standardise procedures. Within the country, ICAR–NBAIR has a long history of supplying trichogrammatids, chrysopids, coccinellids and other parasitoids and predators via courier or postal services. Most of the routine packing procedures have been brought into conformity with set standards in order to ensure consistency. This means that the methods are highly reliable and ensure secure containment and a suitable environment in transit. To determine if the insects/arthropods will survive transit, a test is carried out, whenever feasible, to replicate the type of handling and environmental conditions that packages are likely to encounter in transit [9]. Some of the common procedures for shipping living arthropods are dealt with elsewhere in this issue [10, 11].

Shipment of insects for non-commercial research purposes

Shipment of insects for non-commercial research purposes, including identification or taxonomic research, should comply with the BDA and the Biological Diversity Rules, 2004. In accordance with these rules and regulations, which are primarily intended to protect national biological resources and traditional knowledge, Indian citizens/institutions wishing to send or hand-carry biological resources to institutions abroad must obtain prior approval from the NBA.

Regulations and guidelines on access to biological resources for research purposes

In pursuance of the Nagoya Protocol on Access to Genetic Resources and the Fair and Equitable Sharing of Benefits arising from their Utilisation to the Convention on Biological Diversity, which has been in place since 2014, the regulations and guidelines regarding access to biological resources and associated knowledge shall apply when resources are to be accessed for research purposes. Access to insect resources from India requires the necessary permissions and approvals from the NBA, as stipulated in Chapter II of the BDA, which covers non-commercial research or research for emergency purposes outside India. It reads as follows:

(1) Any Indian researcher/government institution who intends to carry/send the biological resources outside India to undertake basic research other than collaborative research referred to in section 5 of the Act shall apply to the NBA in Form B annexed to these regulations.
(2) Any government institution which intends to send biological resources to carry out certain urgent studies to avert emergencies like epidemics, etc., shall apply in Form B annexed to these regulations.
(3) The NBA shall, on being satisfied with the application under section 3 sub-section 2, accord its approval within a period of 45 days from the date of receipt of the application.
(4) On receipt of approval of the NBA, the applicant shall deposit voucher specimens in the designated national repositories before carrying/sending the biological resources outside India and a copy of proof of such deposits shall be sent to the NBA.

If the research is collaborative, as per Section 5 of the BDA, it is not necessary to obtain approval from the NBA. Instead, live insects for non-commercial research can be exported

with the permission of the Department of Agricultural Research and Education, but even so, the NBA must be informed of the export. The collaborative research proposed should conform to Government of India policy and be approved by the Government of India.

Application process

The application process for export involves submitting specific forms to obtain approval from the NBA [7]. Form B is used by Indian researchers to send or hand-carry insect specimens abroad for identification or taxonomic research purposes. Before insects are dispatched, voucher specimens must be deposited at an appropriate designated repository for future reference. Over the last few years, ICAR–NBAIR scientists have obtained multiple Form B approvals so that specimens can be sent abroad for identification (for non-commercial research) by international scientists. Most of the specimens sent by ICAR–NBAIR for identification are arachnids or species belonging to the orders Coleoptera, Diptera, Hemiptera, Hymenoptera and Thysanoptera. A total of 146 Form Bs were processed and approved by the NBA until 2021 [12].

Designated repositories for insects, mites and spiders

In order to ensure national biological resources are protected and differentiated from exotic species, there must be a coordinated system for providing evidence that species are endemic to India. The ICAR–National Bureau of Agricultural Insect Resources is the institute responsible for collecting, characterising, documenting, conserving, exchanging, researching and utilising agriculturally important insect resources (including mites, spiders and related arthropods) for sustainable agriculture.

In September 2012, the Ministry of Environment and Forests (now the Ministry of Environment, Forest and Climate Change), recognised ICAR–NBAIR (which at that time was the National Bureau of Agriculturally Important Insects) as a designated repository for agriculturally important insects, mites and spiders. Its mandate is to be a depository for the safe storage of specimens (holotypes, isotypes or paratypes) of various insect species of new taxa discovered in India; ICAR–NBAIR also stores voucher samples of biological resources that are sent/carried abroad by Indian citizens/institutions for research. The procedures that repositories must follow when storing samples are described in the BDA, along with Rule 14(6) (viii) of the Biological Diversity Rules, 2004, and clause 4(6) of the Regulation on Access and Benefit Sharing Guidelines published in November 2014 [13]. Earlier, in August 2008, the Zoological Survey of India in Kolkata became a designated repository for fauna (which includes insects and related arthropods) and the Jabalpur-based Tropical Forest Research Institute (under the Indian Council of Forestry Research and Education, Dehradun) was designated a repository for termites, butterflies and moths.

Because of the importance of taxonomic specification for both imports and exports of insect resources, reference collections are essential. The National Insect Museum located at ICAR–NBAIR has over 216,400 dry-mounted specimens, including 375 type specimens, and more than 200,000 specimens in wet preservation. Most of the specimens in the collection are Indian, but there are also specimens of exotic beetles, wasps, flies and moths from various countries and regions, including Argentina, Australia, China, Iran, Japan, Nepal, Reunion Island, the United States of America and the West Indies. The collection also contains many undescribed species from remote islands.

Shipment of insects for commercial purposes

There has been a surge in the use of insects as food and feed in many parts of the world in recent years and a similar increase is expected in India. Entomophagy, the practice of eating insects, is an age-old tradition amongst many people across the globe. More and more ready-to-eat insect-based food formulations are entering the local markets in western countries because of increasing consumer demand and acceptance. Insect-based foods, however, are not included in the global standards for foods set by the Codex Alimentarius.

In India, entomophagy is common in north-eastern areas, where tribal communities consume seasonal borers, silkworms and stink bugs as delicacies. The larvae and pupae of lepidopteran insects are sold throughout the year in local markets for consumption. For example, the fresh larvae and pupae of wild tasar silkworm, *Antheraea* spp. (Lepidoptera: Saturniidae), are sold in local markets in Arunachal Pradesh and Meghalaya, where the insect is also available in powdered form. The practice of rearing eri silkworm, *Samia cynthia ricini* (Lepidoptera: Saturniidae), is widespread in Arunachal Pradesh, Manipur, Meghalaya, Mizoram and Nagaland. Consumption of termite alates is a common practice followed by tribal communities in Tamil Nadu and some parts of Bihar and Uttar Pradesh (Amala *et al*., unpublished data). Indigenous knowledge of the season-specific availability of edible insects is passed from generation to generation.

The emergence and abundance of edible insects is largely season-driven and is directly correlated with the availability of their preferred food plants. Bamboo borers and stink bugs, which are widely consumed in tribal areas of the north-east, are examples of species with limited seasonal emergence. Entomophagy is an established practice in these areas, but in tropical parts of India most people largely avoid consuming

insects. This could be due to health concerns and fears about possible allergenicity, or simply fear of the new or unknown. At this time, there is a lack of robust edible-insect processing systems in place in India to attract consumers across different age groups towards entomophagy.

As well as being a possible source of food for humans, insects, which are high in protein, can partly replace the increasingly expensive protein ingredients of the compound feeds used in the livestock, poultry and aquaculture industries. An added advantage is that, by rearing insects on food waste, low value organic biomass is converted into a valuable commodity [14, 15]. But the concept of using insects as feed is still at an early stage in India. An exception to this is the already widespread use of yellow mealworms, *Tenebrio molitor*, and superworms, *Zophobas morio* (both Coleoptera: Tenebrionidae), as food for birds and other companion animals. Farming of the black soldier fly (BSF), *Hermetia illucens* (Diptera: Stratiomyidae), is at a developmental stage in various parts of India, with a few commercial producers in Andhra Pradesh rearing insects for use as poultry feed. A number of protein-rich feed formulations for fish and poultry are in development, with some already at the pre-commercialisation stage. The ICAR–National Bureau of Agricultural Insect Resources has developed a BSF-based protein-rich fish feed (sinking and floating types) in collaboration with the ICAR–Central Inland Fisheries Research Institute in Barrackpore. Black soldier flies are supplied to domestic destinations based on requirements (Figure 1).

Online marketing sites such as Amazon (https://www.amazon.in), Flipkart (https://www.flipkart.com) and Snapdeal (https://www.snapdeal.com) sell insects (e.g. crickets) and annelids (e.g. tubifex worms and bloodworms) for use as feed for fish, pet birds and reptiles. They are sold on a large scale in India and are available in different formulations, including processed worm powder and whole larvae (mealworms, superworms, BSF larvae). These products are marketed with catchy trade names with the prefix 'proty' or 'pro' to convince customers of their protein-rich nature. Many suppliers claim these products are organically reared and rich in protein, calcium and polyunsaturated fatty acids that support the growth and development of tank-grown fishes. Worms are sold in different forms: live, dehydrated and oven-dried. They are sold in attractive sealed containers to meet customer preference, with labels indicating the proximate composition of the feed in the containers. Live larvae are packed with suitable filler materials to ensure that they are not damaged in transit. Some labels in the market bear the suggested rate of usage of the feed per day, based on the age of the pet birds or fish. Even though the content of the products sold online is not certified or established scientifically and there are no strict guidelines in place, the sale of insect-based pet food is prospering in India. As the market for human and animal feed develops further, India may begin to trade insects internationally.

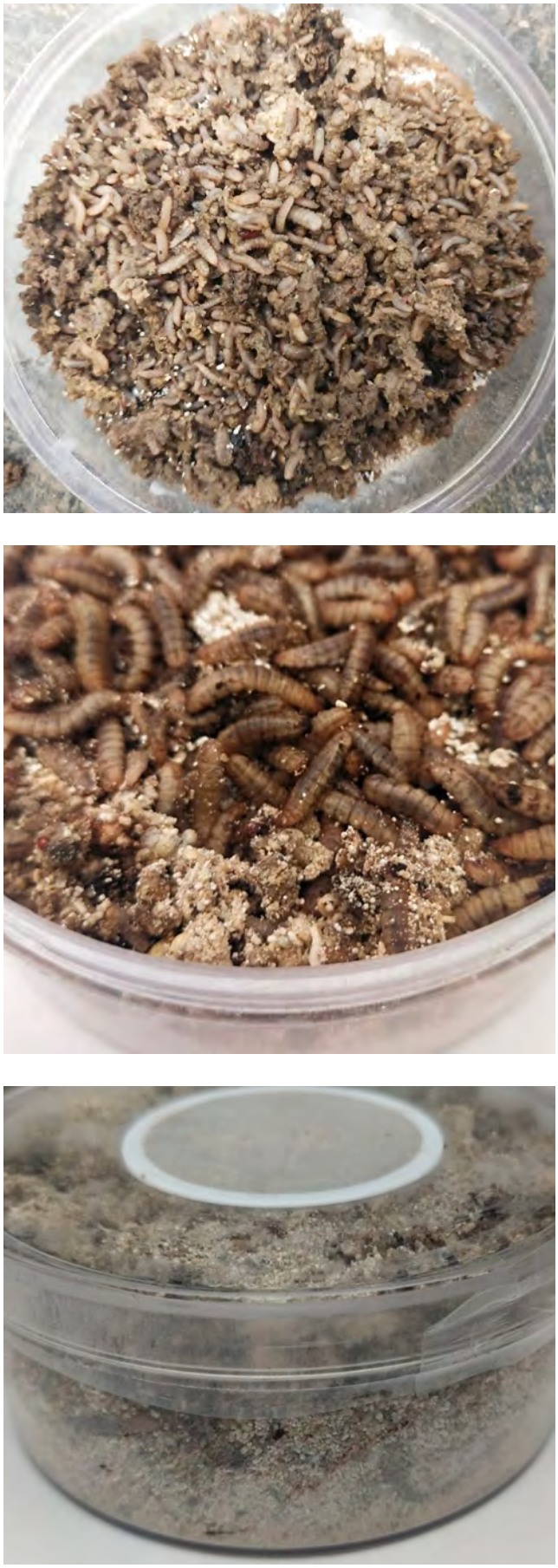

Figure 1

Consignment of live black soldier flies ready for shipment

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Implications, conclusions and recommendations

The procedures and protocols for importing insects and related arthropods for biological control are relatively simple. Until recently, the procedures for transferring insects for non-commercial research were not as straightforward, but the NBA has now introduced a set of regulatory procedures and standard protocols that simplify the dispatch of dead specimens to other countries. Transboundary exchange of insect specimens enables taxonomists to take up exhaustive revisionary, phylogenetic or evolutionary studies on specific insect taxa. With the limited taxonomic expertise available in India, and with restricted access to the type specimens located in museums abroad, documenting species diversity is becoming a herculean task. If specimens are not physically compared with the closely allied species, the chances of misidentifying cryptic taxa are high. This slows down biodiversity documentation, which in turn reduces the trade of potential beneficial species and limits benefit sharing. As for live-insect trade, strict regulations should continue to be enforced, as this will help to prevent the illegal import of commercial insects that may carry pests and/or pathogens that have the potential to cause threats to the native insect biodiversity in India.

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Expéditions d'insectes et autres arthropodes connexes à destination et en provenance de l'Inde à des fins de recherche ou de commerce

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Résumé

Les auteurs présentent succinctement les directives et réglementations régissant les expéditions d'insectes et autres arthropodes connexes à destination et en provenance de l'Inde. Ils examinent le décret de 2003 relatif à la quarantaine végétale (réglementation des importations en Inde), qui couvre également les importations d'agents destinés à la lutte biologique, ainsi que la loi de 2002 sur la diversité biologique et les dispositions réglementaires de 2004 qui lui sont associées concernant les transferts d'insectes et autres arthropodes connexes à des fins d'identification, de recherche taxonomique et de commerce. Le commerce d'insectes destinés à la production de denrées alimentaires ou d'aliments pour animaux est également mentionné. Enfin, les auteurs exposent un certain nombre de conséquences, de conclusions et de recommandations en la matière.

Mots-clés

Agent de lutte biologique – Autorité nationale de la biodiversité – Bioressource – Échange d'insectes – Inde.

Entrada y salida de la India de insectos y otros artrópodos conexos con fines de investigación o comercio

P. Sreerama Kumar, K. Sreedevi, U. Amala, A. Gupta & A. Verghese

Resumen

Los autores resumen sucintamente las directrices y reglas que rigen los envíos de entrada o salida de la India de insectos y otros artrópodos conexos. En este sentido, exponen el Decreto de Cuarentena Vegetal (Reglamento de Importaciones a la India) de 2003, que regula la importación de agentes destinados a actividades de control biológico, y la Ley de Diversidad Biológica de 2002 y el correspondiente Reglamento de Diversidad Biológica de 2004, que rigen

el traslado de insectos y otros artrópodos conexos con fines de identificación, investigación taxonómica o comercio. Tras referirse al comercio de insectos con determinados fines de alimentación humana o animal, acaban exponiendo ciertas consecuencias de todo ello y presentando una serie de conclusiones y recomendaciones al respecto.

Palabras clave

Agente de control biológico – India – Intercambio de insectos – Organismo Nacional de Biodiversidad – Recurso biológico.

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